

On Finiteness and Modal Mismatches under Clausal Ellipsis in Japanese

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Abstract: This paper first makes a novel observation that sluicing in Japanese allows finiteness mismatches. Then, an account of this observation is proposed based on the idea that ellipsis sites can be smaller than usually assumed, which makes it possible to maintain a syntactic identity-based approach to ellipsis mismatches by rendering the mismatches in question as only apparent ones. To be more specific, it is proposed that sluicing can target a ν P serving as a direct complement of C, instead of a TP, and that the problems induced by such a structure are essentially morphophonological so that they can be repaired by ellipsis. Then, it is pointed out that the proposed analysis offers a solution to a puzzle hidden under a widely accepted analysis of Japanese sluicing employing cleft structures, which requires argument ellipsis targeting a CP to take a TP antecedent. The solution crucially assumes that argument ellipsis can target a ν P serving as a clausal argument instead of a full-fledged CP, and it is shown that this assumption receives independent support since it offers a unified account for another novel observation indicating that argument ellipsis targeting clausal arguments is insensitive to the mismatches regarding the modal particles contained in them.

Keywords: *sluicing, argument ellipsis, finiteness, modal particles, repair-by-ellipsis, Japanese*

1. Introduction

Ellipsis requires an identity relation between the ellipsis site and its antecedent, while certain kinds of mismatches are in fact allowed. Let us take a case of sluicing, exemplified by (1). Given the grammaticality of the example, the difference between the indefinite correlate *something* in the antecedent and the trace/lower copy of the wh-remnant in the sluice should be ignorable (throughout this paper, strikethrough is used to indicate the alleged ellipsis sites while their antecedents are put in a box).

- (1) John bought something, but I don't know [what_i [~~he bought t_i~~]].

It is also well known that sluicing does not allow mismatches below the vP-level, for instance voice-mismatch (see Merchant 2013, among many others), but recent studies have shown that mismatches above the vP-level can be ignored (see, for instance, Rudin 2019; Ranero 2021; Anand, Hardt and McCloskey 2022). Some examples suggesting this point are given in (2), which are adapted from Rudin (2019:266). (2a-d) illustrates mismatches in finiteness, tense, modality, and polarity, respectively.

- (2) a. Sally cooks every night; She learned [how [~~to cook~~]] from her father].
b. Your favorite plant is alive, but you can never be sure for [how long [~~your favorite plant will be alive~~]].
c. Sally knows that there is always the potential for awful things to happen, but she doesn't know [when [~~awful things will/might happen~~]].
d. Either turn in your final paper by midnight or explain [why [~~you didn't turn it in by midnight~~]]!

Sluicing in Japanese patterns with that of other languages in that it does not allow voice-mismatch (see, for instance, Sugisaki 2016), but as far as I can tell, it has not been examined whether it allows the mismatches above the vP-level. Against this background, this paper first observes that sluicing in Japanese allows finiteness mismatches between the sluice and its antecedent. Then, I

propose an account of this observation, arguing that it also offers a novel solution to a puzzle hidden under the so-called pseudosluicing analysis of Japanese sluicing (Takahashi 1994; Nishiyama; Whitman and Yi 1996; Merchant 1998; Saito 2004; Takita 2010; Hiraiwa and Ishihara 2012, among others).¹ Second, I expand the data set so as to include the observation that Argument Ellipsis (AE; Oku 1998, *et seq.*) targeting certain types of control CPs allows mismatches with respect to the modal particles included within the relevant CPs. It is then argued that the analysis proposed for the finiteness mismatches can accommodate the modal mismatches in a unified way. At the same time, it is also shown that the analysis concerning the modal mismatches constitutes an independent argument for the solution to the above-mentioned puzzle.

The gist of the observations to be made (concerning both of finiteness and modal mismatches) and the analysis to be proposed is given in (3).²

(3) a. What appears to be observed:

Antecedent: $[_{XP} \dots X \boxed{[_{YP} \dots Y_{[\alpha]} [_{ZP} \dots Z \dots] \dots}] \dots]$

Ellipsis: $[_{XP} \dots X \{ \cancel{[_{YP} \dots Y_{[\beta]} [_{ZP} \dots Z \dots] \dots}] \dots}] \dots]$ (where $\alpha \neq \beta$)

b. What is actually obtained:

Antecedent: $[_{XP} \dots X [_{YP} \dots Y_{[\alpha]} \boxed{[_{ZP} \dots Z \dots]} \dots] \dots]$

Ellipsis: $[_{XP} \dots X \{ \cancel{[_{ZP} \dots Z \dots]} \dots}] \dots]$

The general pattern of the observations is given in (3a), which shows that the YP in the ellipsis site appears to be elided taking the YP in the antecedent, although the values of the feature on the Y heads

¹ As correctly pointed out by an anonymous reviewer, the analysis to be proposed can (and in fact should) cover similar mismatches found in other languages. Although this paper exclusively focuses on the Japanese data for reasons of space, see Takita (in prep.) for an attempt to accommodate the English data in (2) and some related cases.

² This paper remains agnostic about whether ellipsis (be it sluicing or AE) should be implemented in terms of PF-deletion or LF-copying, although ellipsis is indicated by strikethrough as if it involved deletion.

are not identical to each other, notated here as $[\alpha]$ and $[\beta]$, respectively. That is, there is a mismatch between the head Y in the antecedent and its corresponding head Y in the ellipsis site. To accommodate this pattern, this paper proposes that ellipsis in question is legitimate because the ellipsis site (and its corresponding antecedent) can be “smaller” than what is usually assumed, as schematically illustrated in (3b).³ That is, although the head X takes the YP as its complement and its head Y takes the ZP as its complement under normal circumstances, it becomes possible under ellipsis that the X directly takes the ZP as its complement, skipping the YP. Then, the ZP is in fact elided under identity with the ZP in the antecedent. Since the YP, which is the source of the mismatch, is absent in (3b), the ellipsis process in question is in fact free from any mismatches. In other words, I argue that the “mismatches” observed in the pattern (3a) are only apparent.

This paper is organized as follows. In Section 2, it is first observed that sluicing in Japanese allows mismatches with respect to finiteness, providing certain background on Japanese sluicing. Then, Section 3 proposes and elaborates the analysis based on the approach given in (3b), arguing that it can also solve the puzzle concealed under a widely accepted analysis of Japanese sluicing. Section 4 deals with the modal mismatches under AE targeting clausal arguments. Section 5 is a conclusion.

2. Finiteness Mismatches under Sluicing in Japanese

This section is concerned with the finiteness mismatches under sluicing in Japanese. Before jumping into the crucial data, however, Section 2.1 provides some background on sluicing in Japanese, which

³ As an anonymous reviewer correctly points out, the idea of postulating a “smaller” ellipsis site itself is not novel. For instance, Barros, Elliot and Thoms (2014), building upon Merchant’s (2001) semantic identity-based approach, argue that sluicing can target a reduced cleft/copula clause, which can be taken as smaller than a regular clause. Crucially, the analysis to be proposed in this paper differs from theirs in that it maintains syntactic identity.

helps to understand the crucial data to be presented in Section 2.2.

2.1. Background on Sluicing in Japanese

Some typical examples of sluicing in Japanese are given in (4), where the symbol Δ indicates that certain elements are omitted (Inoue 1978; Takahashi 1994).

- (4) a. [Miku-ga dareka-o mita]-rasii-ga, boku-wa [dare-o Δ (da) ka] siranai
 M.-Nom someone-Acc saw-seem-but I-Top who-Acc Cop Q not.know
 ‘It seems [that Miku saw someone], but I don’t know [who Δ].’
- b. Tenma-wa [naze zibun-ga homerareta ka] wakatteiru-ga,
 T.-Top why self-Nom was.praised Q see-but
 Ichi-wa [naze Δ (da) ka] wakatteinai
 I.-top why Cop Q not.see
 ‘Tenma sees [why he was praised], but Ichi doesn’t see [why Δ].’

In (4a), the correlate corresponding to the sluicing remnant is an indefinite expression *dareka(-o)* ‘someone’ while in (4b) it is a wh-phrase *naze* ‘why’.

There are two major approaches to sluicing in Japanese, which are summarized in (5). Although both of them assume that the antecedent is a TP, they diverge in terms of the structure that they assign to the sluice. The approach in (5a) (see Takahashi 1994, among others), which I call a “genuine” sluicing approach, assumes that the derivation involves wh-movement followed by TP-deletion, just like sluicing in English (Ross 1969; Merchant 2001, among others). The other approach is called a “pseudosluicing” approach (cf. Merchant 1998). There are various versions of this approach, and the one given in (5b) is a version proposed by Saito (2004). He argues that sluicing in Japanese is derived from a cleft construction (which is common among the pseudosluicing approach) by applying AE to the CP-subject serving as the presupposition part (see Section 4 for evidence for AE is applicable to

CP arguments).⁴

(5) Antecedent: ... [TP Miku-ga dareka-o mita] ...

a. Ellipsis: ... [CP dare-o₁ [~~TP Miku-ga t₁-mita~~] [C ka]] ...

b. Ellipsis: ... [_{Cleft} [~~CP Op₁ [~~TP Miku-ga t₁-mita~~] no~~]-ga dare-o₁ (da) ka] ...

M.-Nom saw C-Nom who-Acc Cop Q

‘(lit.) ... [who it is [that Miku saw]] ...’

The optional availability of the copula *da* in (4) has been taken as a strong piece of evidence for the pseudosluicing approach, as it is also optional in cleft constructions as shown in (5b) while there is no place for it under the genuine sluicing approach in (5a).

Although examples like in (4) represent sluicing with finite clauses, Takita (2010) observes that Japanese also has sluicing with infinitival clauses, based on examples like (6). The point is that verbs like *mayotteiru* ‘hesitate’ and *kimekaneteiru* ‘cannot decide’ require their embedded clause to be an infinitival control complement. In (6), the marker *-(y)oo* attached to the embedded verbs in the antecedent ensures that the complement clauses are infinitival (see Uchibori 2000; Fujii 2006, among others).⁵

(6) a. Taroo-wa [PRO dono zyaanaru-ni ronbun-o das-oo ka] kimeta-ga,
 T.-Top which journal-to paper-Acc submit-Inf Q decided-but
 Hanako-wa [doko-ni Δ (*da) ka] mayotteiru/kimekaneteiru
 H.-Top where-to Cop Q hesitate/cannot.decide

⁴ Following Saito (2004), this paper assumes an operator movement is involved in cleft (see also Hoji 1990), but nothing hinges up on this assumption so that the entire discussion is consistent with an approach where the focused phrase itself is moved without help of the operator (cf. Hiraiwa and Ishihara 2012).

⁵ In all the relevant examples to be provided in this paper, PRO is used to indicate the existence of the null subject which is obligatorily controlled by the subject of the matrix clause without any theoretical commitment.

‘(lit.) Taroo has decided [to which journal to submit a paper], but Hanako hesitates/cannot decide [to where Δ].’ (based on Takita 2010:157)

- b. Miku-wa [PRO dono hakubutukan-e otoosan-to ik-oo ka] kimeta-ga,
 M.-Top which museum-to father-with go-Inf Q decided-but
 Tenma-wa [dono doobutuen-e Δ (*da) ka] mayotteiru/kimekaneteiru
 T.-Top which zoo-to Cop Q hesitate/cannot.decide
 ‘(lit.) Miku has decided [to which museum to go with her father], but Tenma hesitates/cannot decide [to which zoo Δ].’

As for the sluice, the fact that the copula *da* cannot appear unlike the case in (4) indicates that the embedded clause containing the ellipsis site is also infinitival. This is because the copula cannot appear in the clause embedded under the verbs in question, as shown in (7).

- (7) Taroo-wa [PRO doko-e ik-oo (*da) ka] mayotteiru/kimekaneteiru
 T.-Top where-to go-Inf Cop Q hesitate/cannot.decide
 ‘(lit.) Taroo hesitates/cannot decide [where to go].’ (based on Takita 2010:151)

In fact, the verbs in question can take neither a simple copula clause nor a cleft clause as its complement, as shown by the ungrammaticality of the examples in (8).

- (8) a. *Taroo-wa [sore-ga doko-e da ka] mayotteiru/kimekaneteiru
 T.-Top it-Nom where-to Cop Q hesitate/cannot.decide
 ‘(lit.) Taroo hesitates/cannot decide [where it is].’
 b. *Taroo-wa [[_{CP} ik-u no]-ga doko-e da ka] mayotteiru/kimekaneteiru
 T.-Top go-Pres C-Nom where-to Cop Q hesitate/cannot.decide
 ‘(lit.) Taroo hesitates/cannot decide [where it is [to go]].’ (based on Takita 2010:151)

Takita (2010) argues that this property, taken together with the grammaticality of examples like (6), strongly indicates that sluicing with to infinitival clauses in Japanese instantiates the English-type genuine sluicing structure. That is, the relevant parts of (6b) must be analyzed as shown in (9), which

involves *wh*-movement followed by TP-deletion, because there is no cleft structure available, which would yield a pseudosluicing structure after AE.

(9) Antecedent: ... [CP dono hakubutukan-e₁ [TP PRO otoosan-to t₁ ik-oo] ka] ...

Ellipsis: ... [CP dono doobutuen-e₂ ~~[TP PRO otoosan-to t₂ ik-oo]~~ ka] ...

The discussion so far suggests that the availability of the copula is an important property that distinguishes sluicing in finite clauses from those in infinite clauses. The examples in (10) provide yet another property that supports the relevant distinction. As shown in (10a), Ross (1969) has already pointed out that infinitival clauses cannot host the *wh*-phrase *why*, and this property is also observed for sluicing applied to an infinitival clause. The example in (10b) suggests that Japanese exhibits the same pattern, where the *wh*-phrase *naze* ‘why’ is prohibited in both the antecedent and ellipsis clauses.

(10) a. Bob knows [how/when/where/how long/*why to crane his neck],

but I don’t know [how/when/where/how long/*why Δ]. (based on Ross 1969:271)

b. Miku-wa [doo/itu/dokode/dorekurai/*naze hasir-oo ka] kimeta-ga,

M.-Top how/when/where/how.long/why run-Inf Q decided-but

Tenma-wa [doo/itu/dokode/dorekurai/*naze} Δ ka] mayotteiru/kimekaneteiru

T.-Top how/when/where/how.long/why Q hesitate/cannot.decide

‘Miku has decided [how/when/where/how long/*why to run], but Tenma hesitates

[how/when/where/how long/*why Δ].’

Since *naze* ‘why’ can appear in cases like (4), where both the antecedent and ellipsis clauses are finite, the pattern found in (10b) indicates that verbs like *mayotteiru* ‘hesitate’ do require infinitival complements.⁶

⁶ As correctly pointed out by an anonymous reviewer, the empirical basis of this paper crucially depends on the existence of infinitival clauses in Japanese, following Uchibori (2000) and Fujii (2006). However, one of the hallmarks of infinitival status, namely the lack of tense-alternation (see (13) for a concrete example), is recently challenged by Akuzawa and Kubota 2020, 2021 and Kubota and

2.2. Core Data

Having established that sluicing in Japanese is possible with both finite and infinitival embedded clauses, this section provides the crucial data involving the finiteness mismatches.

The first paradigm is given in (11). The example in (11a) contains an infinitival clause, which sets up the context for (11b-d), where the matrix verb is *siranai* ‘not know’. The contrast between (11b) and (11c), which do not involve ellipsis, indicates that verbs like *siranai* ‘not know’ take a finite (be it present or past) complement but not an infinitival one containing *-(y)oo*.⁷ When sluicing is applied, the example in (11d) is obtained.⁸

- (11) a. Miku-wa [PRO dokoka-e dekake-yoo to] keikakusiteita-ga, ...
M.-Top somewhere-to go.out-Inf C was.planning-but

Akuzawa 2020 (see also a debate by Fujii, Ogawa and Ono 2023 and Akuzawa and Kubota 2023). Although I do not go into any detail of the debate due to space limitations, it is worth emphasizing that the behavior of *naze* ‘why’ found in cases like (10) serves as another hallmark, which is independent from the tense-alternation generalization.

⁷ As shown in (i), verbs like *siranai* ‘not know’ can take a complement containing *-(y)oo* if the infinitival verb is further embedded under another finite verb (I thank an editor of *JEAL* for raising a question concerning this point).

- (i) ... boku-wa [doko-e (kanozyo-ga) [PRO dekake-yoo to] sita ka] siranai
I-Top where-to she-Nom go.out-Inf C did Q not.know
‘... I don’t know [where she tries [to go out]].’

Notice however that the clause embedded under *siranai* ‘not know’ in (i) is not identical to the clause in (11a), because (11a) involves *keikakusiteita* ‘was planning’, not *sita* ‘did’.

⁸ As pointed out by an editor of *JEAL*, (11d) is ambiguous between the intended reading and the one where the whole antecedent in (11a) is supplied. It is true that there is no mismatch regarding finiteness in the latter case, but what is crucial is that (11d) can have the intended reading. See also footnote 10 for a related discussion.

‘Miku was planning [to go out somewhere], but ...’

- b. boku-wa [doko-e (kanozyo-ga) deka-ru/ta ka] siranai
I-Top where-to she-Nom go.out-Pres/Past Q not.know

‘I don’t know [where she goes/went out].’

- c. *boku-wa [doko-e PRO deka-yoo ka] siranai
I-Top where-to go.out-Inf Q not.know

‘I don’t know [where to go out].’

- d. boku-wa [doko-e Δ ka] siranai
I-Top where-to Q not.know

‘I don’t know [where Δ].’ (Δ = she goes/?*went out)

The grammaticality of (11d) with the intended interpretation indicates that sluicing is possible with an infinitival antecedent and a finite sluice.⁹

⁹ An anonymous reviewer points out that (11d) seems to allow the pronominal subject *sore* ‘it’ to appear in the embedded clause, as shown in (i).

- (i) boku-wa [sore-ga doko-e ka] siranai
I-Top it-Nom where-to Q not.know
‘I don’t know [where it is].’

If this type of structure is indeed available for (11d) and the other examples to be presented in the main text, as hinted by the reviewer, the factual claim of this paper that sluicing allows finiteness mismatches is significantly undermined, as the surface string in (11d) can be derived not via sluicing but via *pro*-drop of the subject. Having said this, however, (i) sounds unacceptable (or at least quite awkward) to my ears. To be more specific, the surface string of (i) itself looks grammatical, but when it is anteceded by (11a), it becomes hard to obtain the proper reference of the pronominal subject. I suspect that this is because the intended antecedent of the pronoun is the event which is to be equated with the PP *doko-e* ‘where’, which is already awkward, and furthermore the event is denoted by the embedded clause of (11a), which is irrealis (by being an infinitival) and contains an indefinite *dokoka-e* ‘to somewhere’, so that it is not so compatible with the definite nature of the pronoun *sore* ‘it’.

As indicated by the translation for (11d), the past tense interpretation is not possible (or at least very hard to obtain). However, it can be shown that this is just an illusion by setting up an appropriate context. Let us consider the paradigm in (12), where (12a) is supposed to be followed by (12b-e).

(12) a. Miku-wa [PRO dokoka kaigai-e dekake-yoo to] keikakusiteita-ga, ...

M.-Top somewhere abroad-to go.out-Inf C] was.planning-but

‘Miku was planning [to go out somewhere abroad], but ...’

b. #[doko-e (kanozyo-ga) dekake-ru ka] pasupooto-no kiroku-kara wakatta

where-to she-Nom go.out-Pres Q passport-Gen record-from turned.out

‘it has turned out from the record on the passport [where she goes out].’

Hence, the above-mentioned concern can be circumvented.

At the same time, it is true that there exists a considerable variation among speakers for the acceptability of the clause of the form *sore-ga PP (da)* ‘It is PP’ (see, for instance, Saito 2004). One way of making the empirical generalization of this paper more solid is to employ multiple sluicing. One such example is given in (ii), which is formed by combining (11a) and (11d) with adding instrumental PPs *nanika/nani-de* ‘by something/what’ to them. In this case, having the pronominal subject leads to more severe degradation, presumably because the clause has the form *sore-ga PPI PP2 (da)* ‘It is PP1 PP2’, namely a copula sentence with multiple PPs in the predicate position.

(ii) Miku-wa [PRO dokoka-e nanika-de dekake-yoo to] keikakusiteita-ga,

M.-Top somewhere-to something-by go.out-Inf C] was.planning-but

boku-wa [(**sore-ga*) doko-e nani-de Δ ka] siranai

I-Top it-Nom where-to what-by Q not.know

‘Miku was planning [to go out somewhere by (using) something], but I don’t know [where by what Δ].’

Note that the embedding predicates in (ii) are the same as the ones in (11a) and (11d). Hence, the mismatch in finiteness is inevitably involved, confirming the generalization that sluicing is possible with an infinitival antecedent and a finite sluice. Similar modifications for all the examples to be provided do not affect the acceptability of them, although I keep employing single rather than multiple sluicing for the rest of the paper for the sake of simplicity.

- c. [doko-e (kanozyo-ga) deka-ke-ta ka] pasupooto-no kiroku-kara wakatta
 where-to she-Nom go.out-Past Q passport-Gen record-from turned.out
 ‘it has turned out from the record on the passport [where she went out].’
- d. *[doko-e PRO deka-ke-yoo ka] pasupooto-no kiroku-kara wakatta
 where-to go.out-Inf Q passport-Gen record-from turned.out
 ‘it has turned out from the record on the passport [where to go out].’
- e. [doko-e Δ ka] pasupooto-no kiroku-kara wakatta
 where-to Q passport-Gen record-from turned.out
 ‘it has turned out from the record [where Δ].’

The sentences in (12b-d) involves no ellipsis while sluicing has applied to (12e). (12d) is ungrammatical for the same reason as (11c), as the verb *wakatta* ‘turned out’ is not compatible with a *-(y)oo*-infinitival complement. What is important is the contrast between (12b) and (12c). In (12b), the embedded clause contains the verb in the present tense form, and this sentence is not felicitous because the embedded clause requires that Miku have not gone out abroad yet but the matrix clause forces the event to have already happened. Hence, (12c), which contains a complement clause with the past tense form, is the only acceptable sequel. Then, the grammaticality of (12e) suggests that it is possible to apply sluicing to a past tense finite clause with an infinitival antecedent.¹⁰

¹⁰ If the ellipsis site in (12e) could be supplied with the whole antecedent clause in (12a), which is finite, there would be no intended mismatch. Note however that it yields another infelicitous reading where Miku’s plan itself but not the fact of her going out somewhere abroad is recorded on her passport, which does not happen given the world knowledge. The fact that (12e) does not have such infelicity ensures the mismatch in question. Put differently, if sluicing could not tolerate the mismatch in question, (12e) should have (i) as the sole possible underlying source, which is infelicitous for the reason mentioned above.

(i) ... #[doko-e deka-ke-yoo to keikakusiteita ka] pasupooto-no kiroku-kara wakatta
 where-to go.out-Inf C was.planning Q passport-Gen record-from turned.out

The examples in (11) and (12) show that sluicing of a finite clause, be it present or past, is possible with an infinitival antecedent. The examples in (13) and (14), on the other hand, suggest that the opposite pattern, namely sluicing of an infinitival clause with a finite antecedent is also possible. Let us start with the case where the past tense finite clause is the antecedent. The relevant examples are given in (13), where (13a) sets up the context for (13b-e).

- (13) a. Miku-wa [moo Tenma-ga dokoka-e dekake-ta to] itteita-ga, ...
 M.-Top already T.-Nom somewhere-to go.out-Past C said-but
 ‘Miku said [that Tenma had already gone out somewhere], but ...’
- b. zituwa kare-wa mada [doko-e PRO dekake-yoo ka] kimekaneteiru
 in.fact he-Nom yet where-to go.out-Inf Q cannot.decide
 ‘in fact, he hasn’t decided [where to go out].’
- c. zituwa kare-wa mada [doko-e (*kare-ga) dekake-ru ka] kimekaneteiru
 in.fact he-Nom yet where-to he-Nom go.out-Inf Q cannot.decide
 ‘in fact, he hasn’t decided [where he goes out].’
- d. *zituwa kare-wa mada [doko-e (kare-ga) dekake-ta ka] kimekaneteiru
 in.fact he-Nom yet where-to he-Nom go.out-Past Q cannot.decide
 ‘in fact, he hasn’t decided [where he went out].’

‘... it has turned out from the record [where she was planning to go out].’

Related to this point, an anonymous reviewer points out that (i) does not sound infelicitous to him/her. I suspect that this discrepancy arises from the following interpretation (possibly obtained via pragmatic enrichment) of the expression *pasupooto-no kiroku-kara wakatta* ‘(it has) turned out from the passport’: Suppose that there is a competent immigrant inspector, who is so experienced that the inspector can predict where a person would plan to go next just inspecting the visas and stamps on the person’s passport. In this situation, I agree with the reviewer in that (i) (and (12e) with this interpretation) is felicitous, but the point is that (12e) as it is does not have this interpretation and is acceptable with the intended reading requiring the mismatch.

e. zituwa kare-wa mada [doko-e Δ ka] kimekaneteiru

in.fact he-Nom yet where-to Q cannot.decide

‘in fact, he hasn’t decided [where Δ].’

Note that the matrix verb in (13b-e) is *kimekaneteiru* ‘cannot decide’, which exclusively requires a control complement (see (6)-(8)). This requirement is straightforwardly satisfied in (13b), where the embedded verb is accompanied with *-(y)oo*. As for (13c), although the morpheme *-ru* appears on the embedded verb, the fact that an overt subject is not possible indicates that the clause is an infinitival one (hence the morpheme is glossed as Inf but not Pres). In fact, Fujii (2006) argues that the infinitival verbs in Japanese may assume a finite form, especially when the verb in question is not allowed to alternate between the present tense form and the past tense form. For the case at hand, the ungrammaticality of (13d), which is minimally different from (13c) in terms of the embedded verb forms, shows that the alternation is prohibited. Hence, the requirement of the matrix verb is satisfied for (13c) as well. The contrast between (13b-c) and (13d) thus suggests that the embedded clause identical to the one in the antecedent (13a) results in ungrammaticality in this context. Therefore, the grammaticality of (13e), to which sluicing has applied, indicates that sluicing is applicable to an infinitival clause with a past tense finite clause antecedent.¹¹

Similarly, the paradigm in (14) shows that sluicing of an infinitival clause is possible with a present tense finite clause antecedent. In (14a), which precedes (14b-e), the embedded verb is accompanied with the morpheme *-ru*. However, it is unambiguously an instance of the present tense marker, because the embedded clause has the overt subject *Tenma(-ga)*, which is different from the

¹¹ An anonymous reviewer asks whether (13e) (as well as (14e)) can have the pronoun *sore* ‘it’ as the embedded subject (see also footnote 9). I do not find the modified versions of (13e) and (14e) with *sore* acceptable, and this is presumably because the embedding verbs in the relevant examples are *mayotteiru* ‘hesitate’ and *kimekaneteiru* ‘cannot decide’, which cannot take a copula/cleft sentence as its complement (see (8)).

matrix overt subject *Miku(-wa)*.¹² On the other hand, the one in (14c) is the infinitive marker, as the overt subject is not allowed just like (13c).

- (14) a. Miku-wa [tikazika Tenma-ga dokoka-e dekake-ru to] itteita-ga, ...
 M.-Top near.future T.-Nom somewhere-to go.out-Pres C said-but
 ‘Miku said [that Tenma would go out somewhere in the near future], but ...’
- b. zituwa kare-wa mada [doko-e PRO dekake-yoo ka] kimekaneteiru
 in.fact he-Nom yet where-to go.out-Inf Q cannot.decide
 ‘in fact he hasn’t decided [where to go out].’
- c. zituwa kare-wa mada [doko-e (*kare-ga) dekake-ru ka] kimekaneteiru
 in.fact he-Nom yet where-to he-Nom go.out-Inf Q cannot.decide
 ‘in fact he hasn’t decided [where he goes out].’
- d. *zituwa kare-wa mada [doko-e (kare-ga) dekake-ta ka] kimekaneteiru
 in.fact he-Nom yet where-to he-Nom go.out-Past Q cannot.decide
 ‘in fact he hasn’t decided [where he went out].’
- e. zituwa kare-wa mada [doko-e Δ ka] kimekaneteiru
 in.fact he-Nom yet where-to Q cannot.decide
 ‘in fact he hasn’t decided [where Δ].’

This observation, taken together with the fact that neither the *-(y)oo*-infinitival complement in (14b) nor the past tense finite complement in (14d) is identical to the one in (14a), suggests that no identical antecedent is available for the ellipsis site in (14e). The grammaticality of (14e) then indicates that such a mismatch can be ignored.

¹² An editor of *JEAL* notes that the morpheme *-ru* in (14a) might be better called the *non-past* rather than *present* tense marker (a similar comment is made by another editor). I agree with them, but the point being made here is that it instantiates the finite tense property, which is for instance evidenced by the availability of nominative subjects. Hence, I keep using the term present tense for simplicity.

Finally, let us consider whether the observations made so far are consistent with the behavior of the copula *da* mentioned in Section 2.1. The examples in (15) are constructed from the ones in (11), (12), (13) and (14), respectively, by picking up only the antecedent and the sluice and combining them.

- (15) a. Miku-wa [PRO dokoka-e dekake-yoo to] keikakusiteita-ga,
M.-Top somewhere-to go.out-Inf C was.planning-but
boku-wa [doko-e Δ (da) ka] siranai
I-Top where-to Cop Q not.know
‘Miku was planning [to go out somewhere], but I don’t know [where Δ].’ (Δ = she goes out)
- b. Miku-wa [PRO dokoka kaigai-e dekake-yoo to] keikakusiteita-ga,
M.-Top somewhere abroad-to go.out-Inf C] was.planning-but
[doko-e Δ (*da) ka] pasupooto-no kiroku-kara wakatta
where-to Cop Q passport-Gen record-from turned.out
‘Miku was planning [to go out somewhere abroad], but it has turned out from the record [where Δ]. (Δ = she went out)’
- c. Miku-wa [moo Tenma-ga dokoka-e dekake-ta to] itteita-ga,
M.-Top already T.-Nom somewhere-to go.out-Past C said-but
zituwa kare-wa mada [doko-e Δ (*da) ka] kimekaneteiru
in.fact he-Nom yet where-to Cop Q cannot.decide
‘Miku said [that Tenma had already gone out somewhere], but in fact, he hasn’t decided [where Δ]. (Δ = to go out)’
- d. Miku-wa [tikazika Tenma-ga dokoka-e dekake-ru to] itteita-ga,
M.-Top near.future T.-Nom somewhere-to go.out-Pres C said-but
zituwa kare-wa mada [doko-e Δ (*da) ka] kimekaneteiru
in.fact he-Nom yet where-to Cop Q cannot.decide
‘Miku said [that Tenma would go out somewhere in the near future], but in fact he hasn’t

decided [where Δ]. (Δ = to go out)'

Putting aside (15b) for a moment, the patterns found in the example in (15a) and the ones in (15c-d) indicate that the copula is optionally allowed only in the former. This is consistent with the claim that the sluice in (15a) is a finite clause while the one in (15c-d) is an infinitival, as is expected from the choice of the matrix verb. This in turn confirms that there is indeed a mismatch between the sluice and its antecedent in the relevant examples in (15).

Returning to (15b), the copula is not allowed in this case despite the fact that the matrix verb can in principle take a past tense embedded clause.¹³ In other words, the sluice allows the intended past tense interpretation only when the copula is absent. In the next section, I illustrate how the idea sketched in Section 1 accommodates all the observations made in this section, including the rather mysterious behavior of the copula found in (15).

3. Proposals and Analysis

This section aims at providing an analysis of the observations made in the previous section. As a first step, let us schematically summarize the observations made in Section 2.2 in terms of the approaches to sluicing reviewed in Section 2.1. The structure in (16) corresponds to the case where the antecedent is an infinitival clause and the sluice is a finite clause, while the one in (17) schematizes the opposite case.

(16) Antecedent: ... [CP [TP ... correlate ... V-v-T_[Inf]] C] ...

a. Ellipsis: ... [CP wh₁ [TP ... t₁ ... V-v-T_[Fin]] C_Q] ...

b. Ellipsis: ... [Cleft [CP ~~Op~~₁ [TP ... t₁ ... V-v-T_[Fin]] C] wh₁ (Cop) C_Q] ... (* if [Fin] = Past)

¹³ To be more precise, (15b) with the copula seems to yield only the infelicitous (and irrelevant) reading where Miku's plan, but not her going out, is recorded on her passport, which is obtained by supplying the whole finite antecedent clause to the ellipsis site, instead of the intended infinitival embedded clause (see also footnote 10).

(17) Antecedent: ... [CP $\boxed{[TP \dots \text{correlate} \dots V\text{-}v\text{-}T_{[Fin]}]}$ C] ...

Ellipsis: ... [CP wh₁ [$\overline{[TP \dots t_i \dots V\text{-}v\text{-}T_{[Inf]}]}$] CQ] ...

Recall that sluicing with finite clauses may have a genuine sluicing structure like (16a) or a pseudosluicing structure like (16b). For the sake of discussion, let us assume that sluicing with the overt copula unambiguously has the structure in (16b) while the one without it is structurally ambiguous between (16a) and (16b). Given that the copula is allowed only when sluicing applies to a present tense clause, as shown by the contrast between (15a) and (15b), it follows that sluicing with a present tense clause can have a genuine sluicing structure or a pseudosluicing structure while that with a past tense clause is compatible only with the former. As for sluicing with infinitival clauses, the structure is unambiguously the genuine sluicing one as discussed in Section 2.1, so the structure in (17) is the only possibility. This is also consistent with the behavior of the copula found in (15c-d).

Now, the point is that no matter what kind of structure is assumed, it is clear that in (16) and (17) there is a mismatch between the sluice and the antecedent with respect to the feature value concerning finiteness, namely [Fin(ite)] vs. [Inf(initive)]. In what follows, I first provide an account for the cases in (16a) and (17), namely the ones with genuine sluicing structures. Then, I come back to the case with a pseudosluicing structure in (16b), as the latter is a bit more complicated than the former.

Let us start discussion with the cases in (16a) and (17). Notice that the configurations in question instantiate the situation mentioned in (3a). Under the standard assumption, a CP selects a TP, which in turn selects a vP, so that the presence of T is inevitable. Since T must be present in the structure, the mismatch found in (16)-(17) seems inevitable as well. Following the general idea sketched in (3b), however, this paper proposes to abandon the assumption mentioned above and argues for the ideas in (18).

(18) a. C in the sluice may directly select vP skipping TP under ellipsis, and the vP can take the identical vP in the antecedent.

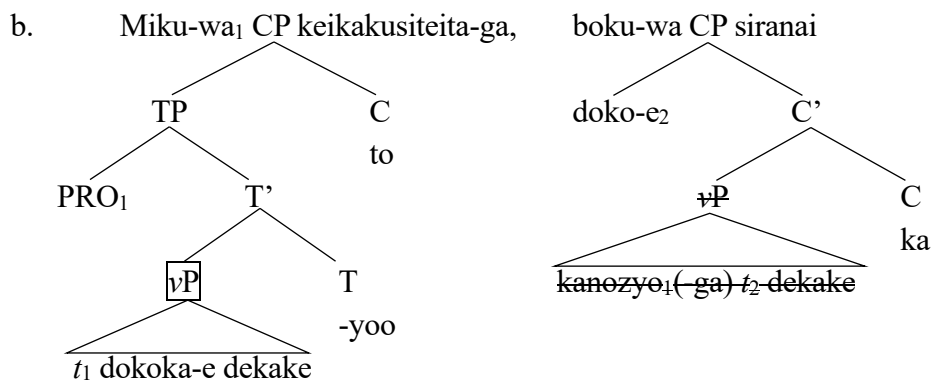
b. The problems caused by such an unusual complementation relation can be “repaired” by

ellipsis.

Before providing some theoretical and conceptual justifications for (18), I illustrate how the proposed analysis accounts for the relevant data. Taking (15a) without the copula as a concrete example, the proposed analysis assigns it the structure depicted in (19). In (19a), the antecedent has a usual structure where the complementizer *to* ‘that’ takes as its complement the TP headed by the infinitive marker *-(y)oo*, which in turn selects the *vP*. On the other hand, the sluice in (19b) has a rather unusual structure: The interrogative *C ka* directly takes the bare *vP* as its complement, and sluicing applies to this *vP*, taking another *vP* in the antecedent.¹⁴ The same point is illustrated in (19b) in terms of tree diagrams, which makes it visually clear that the antecedent has a run-of-the-mill CP-TP-*vP* structure while the sluice has the CP-*vP* structure.

- (19) a. Miku-wa₁ [CP [TP PRO₁ [vP t₁ dokoka-e dekake] [T -yoo]] [C to]] keikakusiteita-ga,
M.-Top somewhere-to go.out -Inf C was.planning-but
boku-wa [CP doko-e₂ [~~vP kanozzyo-ga₁ t₂ dekake~~] [C ka]] siranai
I-Top where-to she-Nom go.out Q not.know
‘Miku was planning [to go out somewhere], but I don’t know [where Δ].’

¹⁴ An anonymous reviewer wonders whether the proposed analysis implies that Japanese has *vP/VP*-ellipsis. Although this is an interesting and important question, some clarification seems necessary. Note that what is proposed in this paper is that the target of sluicing, licensed by *C*, can be a *vP*. In this sense, the answer is yes; Japanese does have ellipsis of a *vP*. This, however, does not necessarily imply that Japanese has *vP/VP*-ellipsis, construed as referring to the case where ellipsis in question is licensed by *T* (i.e., cases like “Can you eat natto? --- Yes, I can [~~vP/VP eat natto~~]!”). Although the paper remains agnostic about the status of *vP/VP*-ellipsis in the latter sense, it is important to explore whether there are other cases where a *vP* is targeted by ellipsis. In fact, I argue in later in this section and Section 4 that this is indeed the case, as AE can also target a *vP* serving as an argument.



Under this analysis, there is in fact no mismatch between the sluice and its antecedent, because the TP-level projection, whose head is the source of the mismatch, is present only in the antecedent and it is located outside of the vP .¹⁵ That is, under the proposed analysis, the mismatches observed in Section 2.2 are in fact an illusion. Put differently, the proposal in (18a) allows us to avoid the mismatches by discarding the standard assumption concerning the CP-TP- vP relation.

¹⁵ An anonymous reviewer correctly points out that this analysis may predict that “TP-adverbs” cannot appear in the sluice due to the absence of the TP-projection. This prediction seems to be incompatible with the observation made by the reviewer in (i) with a slight modification, where the temporal adverb *asita* ‘tomorrow’ is included in the embedded clause of the sluice.

- (i) Miku-wa [asita-kara mainiti dokoka-e de kake-yoo to] keikakusiteita-ga,
M.-Top tomorrow-from everyday somewhere-to go.out-Inf C was.planning-but
boku-wa [asita doko-e Δ ka] siranai.
I-Top tomorrow where-to Q not.know

‘Miku was planning [to go out somewhere from tomorrow to everyday], but I don’t know [where Δ tomorrow].’

However, the problem disappears, given the assumption to be made in the subsequent discussion in the main text that C originally bears the information regarding finiteness and tense and passes it down to T if present. That is, in the case where the TP-level projection is absent, C retains the relevant information that suffices to license the temporal adverbs. Meanwhile, it is noteworthy that the proposed analysis does predict that the proposed CP- vP structure fails to accommodate the phrases that specifically requires a TP as a category, but not just tense information, as its host. I leave it for future research if and how this prediction can be verified.

For the sake of completeness, (20) illustrates the analysis of the case with the infinitival-antecedent/finite-sluiice pair (see (15c-d)). No matter whether the antecedent is in the past tense or in the present tense, no problem arises in (20), because the sluice does not contain the TP-projection, and what is elided is the vP, which is a direct complement of the interrogative C.

- (20) Miku-wa [CP [TP Tenma-ga₁ [[vP t₁ dokoka-e dekake]] [T -ta/ru]] [C to]] itteita-ga,
M.-Top T.-Nom somewhere-to go.out -Past/Pres C has.said-but
kare-wa₁ [CP doko-e₂ [[vP ~~PRO~~₁ ~~t₂~~ ~~dekake~~]] [C ka]] kimekaneteiru
he-Top where-to go.out Q cannot.decide
‘Miku has said that Tenma went/goes out somewhere, but he hasn’t decided [where Δ].’

In this way, the relevant portion of the observations made in Section 2 can be captured by the analysis maintaining a syntactic identity condition between an ellipsis site and its antecedent.

What is crucial for the proposed analysis is that a C head can directly take a vP complement in the absence of T. However, this seems to require ellipsis to apply, as explicitly stated by (18a). In fact, it is evident from the ungrammaticality of the examples in (21), which are obtained by overtly realizing the ellipsis sites of the second clause of (19) and (20), respectively.

- (21) a. ... *boku-wa [CP doko-e₂ [vP kanozyo-ga₁ t₂ dekake] [C ka]] siranai
I-Top where-to she-Nom go.out Q not.know
‘(lit.) ... I don’t know [CP where [vP she go out]].’
b. ... *kare-wa₁ [CP doko-e₂ [vP PRO₁ t₂ dekake] [C ka]] kimekaneteiru
he-Top where-to go.out Q cannot.decide
‘(lit.) ... he hasn’t decided [CP where [vP PRO go out]].’

As for this problem, the proposal in (18b) comes in. That is, I argue that the source of the ungrammaticality of examples like (21) are essentially morphophonological so that they can be overcome by eliding the problematic portion of the structure.

To be more specific, I claim that there are at least three reasons why the direct complementation

of vP by C induces ungrammaticality, which are listed in (22).

(22) a. (Case-)Licensing of the subject

b. Verbal morphology

c. Labeling (cf. Chomsky 2013, 2015, *et seq.*)

To see the point, let me first illustrate how these problems are circumvented under the CP-TP- vP structure, whose schematic derivation is depicted in (23). At the step in (23a), Merge forms a syntactic object (SO) α , applying to the vP and an external argument, which is either of an NP with its Case-feature unvalued (notated as $NP_{[Case: _]}$) when the clause is a finite one or PRO when it is an infinitival one. Then, the SO β is formed at the step in (23b), where T is introduced to the structure and the NP/PRO undergoes raising, which eventually allows α to be labeled as vP . For the reasons to be clarified (see also footnote 15), I assume that the head T comes with its finiteness unspecified and the value is inherited from C (Chomsky 2008, *et seq.*) when it is Merged as in (23c). Once T inherits the value, it becomes possible for β to be labeled as TP, accompanied with the Case-licensing of the NP by $T_{[Fin: Pres/Past]}$ or the licensing of PRO by $T_{[Fin: Inf]}$, as depicted in (23d).¹⁶

(23) a. [α $NP_{[Case: _]}/PRO$ [$vP \dots V-v$]]

b. [β $NP_{[Case: _]}/PRO$ [$vP(=\alpha)$ t_1 [$vP \dots V-v$]] $T_{[Fin: _]}$]

c. [CP [β $NP_{[Case: _]}/PRO$ [vP t_1 [$vP \dots V-v$]]] $T_{[Fin: Pres/Past/Inf]}$ $C_{[Pres/Past/Inf]}$]

d. [CP [$TP(=\beta)$ $NP_{[Case: Nom]}/PRO$ [vP t_1 [$vP \dots V-v$]]] $T_{[Fin: Pres/Past/Inf]}$ C]

Notice that all the potential problems listed in (22) are circumvented at the final step in (23d). The NP/PRO is successfully (Case-)licensed by the appropriate T . As for the verbal morphology, the sequence $V-v-T$ can be realized as $V-v-ru/ta/(y)oo$ so that no verb stems, which are bound morphemes

¹⁶ Since what is important is that β is successfully labeled, it makes no difference even when we assume that the label of β is in fact $\langle \phi, \phi \rangle$. This labeling results from the assumption that the licensing of Nominative Case involves ϕ -feature agreement (Chomsky 2013, 2015), which may not apply to Japanese. See Saito (2016, *et seq.*) for how labeling succeeds in the absence of ϕ -feature agreement.

in Japanese, are stranded. Finally, no labeling problem arises as α and β are successfully labeled.

Now let us consider how the CP- ν P structure is constructed under the assumptions laid out above. Starting from the same step as the one in (23a), repeated as (24a), C is directly Merged with the SO α at the step in (24b), which I claim underlie the relevant parts of (21) (omitting wh-movement).

- (24) a. [α NP<sub>[Case: __]/PRO [ν P ... V- ν]]
 b. [CP [α NP_{[Case: __]/PRO [ν P ... V- ν]] C_[Pres/Past/Inf]]}</sub>

The problems that the structure in (24b) causes are clear: Due to the absence of T, neither of the Case feature on the NP nor PRO are to able be licensed, the verb stem remains in the form V- ν , and the label of α left undetermined; hence the ungrammaticality of the examples in (21). As for the first two problems, it is easy to see that the problems can be removed by ellipsis, as shown in (25).¹⁷

- (25) [CP [~~α NP_{[Case: __]/PRO [ν P ... V- ν]]}~~ C_[Pres/Past/Inf]]

As for the labeling problems, here induced by α , I argue that they are essentially PF/linearization-related, following Takita (2020). To be more specific, Takita (2020) argues that only labeled SOs are subject to linearization so that the unlabeled SO of the form {NP, ν P} (namely the one where an external argument is Merged) causes a problem for PF because the system cannot determine whether the NP precedes the ν P or vice versa. As for the interpretation at the LF-interface, where unlabeled SOs are assumed to induce a problem under Chomsky's (2013, 2015) framework, Takita (2020) argues no problem arises to begin with given the standard compositional semantics. For the case at

¹⁷ I assume a version of the traditional Case-Filter which implies that Case is a PF-requirement (see Lasnik 2008 for a historical overview). In this sense, whether PRO causes the same problem as overt NPs in (24b) in fact depends on the assumption regarding how to license PRO. A similar consideration applies to the case where the subject is *pro*. Note however that the problem concerning the verbal morphology still arises even in cases with PRO/*pro*, which requires circumvention by ellipsis. See Lasnik (2001) and references cited therein for some classic cases where morphological problems are salvaged by ellipsis (see also Mendes 2020 and Mendes and Nevins 2023 for a more recent discussion dealing with a wider range of phenomena).

hand, the SO $\{\text{NP}, \text{vP}\}$ can yield a legitimate semantic interpretation with the former being the argument for the function denoted by the latter.

Returning to the labeling problem arising from the SO α in (23)/(24), one way of resolving it is to move the NP as in (23b). This solution is essentially the same with the one proposed by Chomsky (2013, 2015), but the reason why the problem disappears is different; under Takita’s (2020) conception, the SO $\{t_{\text{NP}}, \text{vP}\}$ is still left unlabeled, but it induces no PF/linearization problem because one of the two competing elements, namely the NP, loses its phonological realization. This allows the relative ordering between (the trace/lower copy of) the NP and the vP to be left unspecified without any problem. This in turn opens up another way of resolving the problem: When ellipsis applies as in (25), the necessity of determining the relative ordering of the NP and the vP is lifted up, since neither of them is overtly realized, which allows the label of α to be undetermined.¹⁸

In this way, all the problems arising from the CP-vP structure listed in (22) are “repaired”, or more precisely circumvented, by ellipsis applied in the way depicted in (25). Note at the same time that the structure in (25) correctly yields the desirable semantic interpretations even when the TP-level projection is absent, because it is C by assumption that originally bears information regarding finiteness and tense. Recall that sluices in mismatching cases observed in Section 2.2 have either of the present, past, or infinitival interpretations just like those in non-mismatching cases, depending on the context where they are embedded. Suppose then that the finiteness/tense information exclusively

¹⁸ Under this view, labeling problems never arise for the vP with PRO/*pro*-subjects, as they do not have to be linearly ordered with vP. However, the problem concerning the verbal morphology still remains for (23)/(24) in Japanese, which is an agglutinative language (see also footnote 17). It is then predicted that some languages may allow the CP-vP structure with PRO/*pro*-subjects even without ellipsis if verbal morphology of the languages in question causes no problem (one potential test case is English, where verbal stems can stand alone in certain infinitival clauses). Since exploring this possibility is out of the scope of the current paper, I leave it for future work (but see Takita 2024 for a related discussion).

appears on T, contrary to what is assumed in this paper. Although sluices in non-mismatching cases can have their TPs elided so that no problem should arise, the observations concerning the mismatching cases remains unexplained. For instance, in the case of the infinitival-antecedent/finite-sluice with the past tense interpretation, it is not clear where the past tense interpretation comes from given the general recoverability condition. However, once it is assumed that C bears the information in question (and T inherits it if present), the problem disappears. This assumption is also consistent with the fact that whether the complement clause can be finite or infinitival depends on the matrix verb. For instance, the verb *mayotteiru* ‘hesitate’ exclusively takes infinitival complements. This relationship is naturally captured if it is the embedded C but not the embedded T that must match the matrix verb.

Having illustrated how the proposed analysis accommodates the cases in (16a) and (17), which involve a genuine sluicing structure, let us now turn to the pseudosluicing case represented in (16b), repeated as (26).

(26) Antecedent: ... [CP TP ... correlate ... V-v-T_[Inf] C] ...

Ellipsis: ... [Cleft [_{CP} ~~Op_i [_{TP} ... ~~t_i ... V-v-T_[Fin]~~]~~ C] wh_i (Cop) C_Q] ... (* if [Fin] = Past)

The question is why AE can overcome the mismatch between the infinitival antecedent and the finite sluice. However, if we look at the structure in (26) more closely, it becomes clear that there is another type of mismatch hidden there. That is, the postulated instance of AE in fact applies to the CP-subject taking the TP-antecedent. Moreover, this mismatch is found not only in the cases of the finiteness mismatches but also in run-of-the-mill examples of sluicing without them such as (27a), which is repeated from (4a) with a slight modification. As shown in (27b), although in this case the sluice matches its antecedent in finiteness (and tense), the CP is assumed to be elided under identity with the TP.¹⁹

¹⁹ The assumption that the CP but not the TP serves as the antecedent does not solve the problem, because there arises another mismatch concerning complementizers, namely *to* in the antecedent and *no* in the

- (27) a. Kimi-wa [CP [TP Miku-ga dareka-o mita] [C to]] itteita-ga,
 you-Top M.-Nom someone-Acc saw C said-but
 boku-wa [Cleft [CP Op_i [TP Miku-ga ~~t_i~~ mita] [C no]]-ga dare-o_i (da) ka] siranai
 I-Top M.-Nom saw C-Nom who-Acc Cop Q not.know
 ‘You said [that Miku saw someone], but I don’t know [who it is ~~that Miku saw~~].’
- b. Antecedent: ... [CP [TP ... correlate ... V-v-T_[Fin]] C] ...
 Ellipsis: ... [Cleft [CP Op_i [TP ... ~~t_i~~ ... V-v-T_[Fin]] C] wh_i (Cop) C_Q] ...

That is, the pseudosluicing analysis inherently involves a category-mismatch.

This fact can be taken as an argument for a certain kind of semantic identity approach (notably the one by Merchant 2001). In fact, Potsdam (2007) argues for such an approach based on the data from Malagasy, which lacks the English-type *wh*-movement and only utilizes a cleft strategy to derive *wh*-questions and hence sluicing in the language (see also van Craenenbroeck 2010 for a similar argument based on a particular type of sluicing constructions in Dutch). In the rest of this section, however, I provide a syntactic identity-based approach to the problem, which at the same time covers the cases with the finiteness mismatches.²⁰

In particular, I argue that the example in (27a) actually has a structure depicted in (28a), where a *vP* but not a full-fledged CP serves as the subject of the cleft sentence. This *vP*, being the subject of the cleft sentence, can be subject to AE, I claim, under identity with the identical *vP* in the antecedent.²¹

ellipsis site.

²⁰ It is thus expected that the general approach of this paper can also accommodate the Malagasy/Dutch-type data. If this expectation can be fulfilled, it follows that the data in question do not necessarily constitute strong arguments for the semantic identity-based approach, to which I hope to return in future work.

²¹ Although it is omitted in the illustration in (28), the antecedent *vP* can be made more strictly identical to the elided *vP* if existential closure applies at the *vP* level for the cases with indefinite correlates or successive-cyclic *wh*-movement takes place for the cases with *wh*-correlates as shown in (i).

- (28) a. Kimi-wa [CP [TP Miku-ga₁ [vP t₁ dareka-o mi]] [T -ta]] [C to]] itteita-ga,
 you-Top M.-Nom someone-Acc see -Past C said-but
 boku-wa [Cleft [vP Op₂ [vP Miku-t₂ mi]] dare-o₂ (da) ka] siranai
 I-Top M. see who-Acc Cop Q not.know
 ‘You said [that Miku saw someone], but I don’t know [Cleft who it is [vP Miku see]].’
- b. Antecedent: ... [CP [TP ... [vP ... correlate ... V-v]]-T_[Fin]] C] ...
 Ellipsis: ... [Cleft [vP Op₁ [vP ... t₁ ... V-v]]] wh₁ (Cop) C_Q] ...

All the problems arising from the bare vP lacking the C-T combination (cf. (22)) are successfully circumvented in the same way as the case of genuine sluicing structures, but it is AE that has applied. Since the problems in question are PF-related, the difference between the specific operations involved in these two cases do not make any difference with respect to the repair effect.

Although the case (28b) is an instance of the configuration mentioned in (3b), it has another difference from the genuine sluicing cases. That is, in this case not only the TP but also the CP is missing, as the vP directly serves as the subject of the cleft sentence. This difference in fact paves the way for accounting the rather puzzling behavior of the copula *da* found in (15b), repeated as (29).

- (29) Miku-wa [PRO dokoka kaigai-e deka-ke-yoo to] keikakusiteita-ga,
 M.-Top somewhere abroad-to go.out-Inf C] was.planning-but
 [doko-e Δ (*da) ka] pasupooto-no kiroku-kara wakatta
 where-to Cop Q passport-Gen record-from turned.out
 ‘Miku was planning [to go out somewhere abroad], but it has turned out from the record [where Δ].’

The puzzle is that in (29), which is an instance of the infinitival-antecedent/finite-slucice pair, the copula cannot appear despite the fact the matrix verb is compatible with it. Put differently, the pattern

(i) Antecedent: ... [CP [TP ... [vP Op/wh₁ [vP ... correlate_{Indef}/t₁ ... V-v]]]-T_[Fin]] C] ...

in (29) suggests that the past tense interpretation is blocked when the copula appears.

Recall that the relevant part of the sluice in (29) without the copula can have a genuine sluicing structure with the CP- ν P structure partially depicted in (30a).²² Given the assumption that the finiteness and tense information originates in C, this structure can yield the past tense interpretation if the C has the relevant value. On the other hand, (30b) illustrates the pseudosluicing structure of the relevant part of the sluice in (29). Since the C as well as the T is missing in the presuppositional CP-subject of the cleft clause, and the antecedent is infinitival, there is no way to recover the past tense information for the verb in the presupposition part (namely *dekake* ‘go out’ for the case at hand), although the application of AE itself is legitimate.²³

(30) a. ... [CP doko-e₁ [ν P-~~kanozyo~~ ~~t₁~~-~~dekake~~] [C ka_[Past]]] ...

b. ... [Cleft [ν P-Op [ν P-~~kanozyo~~ ~~t₁~~-~~dekake~~]] doko-e₁ (da) [C ka]] ...

Given that (30b) is the only way to derive (29) with the overt copula, the observation that it is not allowed in this case can be explained as follows: The matrix clause forces the past tense interpretation for the embedded clause but it can only express a ν P-level interpretation which lacks any tense specification.²⁴ This problem cannot be repaired by ellipsis, as it is not morphophonological but semantic in nature.

To summarize this section, it is proposed that the observed patterns concerning the finiteness

²² For the ease of simplicity, the unvalued Case-feature on the subject is omitted. The label “ ν P” is also used for the sake of illustration, as the SO in fact remains unlabeled given the assumptions made earlier.

²³ As mentioned in footnote 21, the antecedent for the ν P containing Op in (30b) can also be made to host an operator.

²⁴ Note that the embedded cleft clause does have C, which can have the past tense information, but it is to be associated with the copula *da* (in such case it assumes the past tense form *datta*). Since the tense of the copula is independent from that of the presupposition CP-subject, there is no past tense information available for the verb in the presupposition in (30b).

mismatches found in sluicing in Japanese can be explained by the general idea given in (3b), elaborated as (18), which states that ellipsis sites can be smaller than what is usually assumed and that the problems caused by having such a smaller complement can be repaired by ellipsis if they are PF-related. As for the cases that can have a genuine sluicing structure, it is argued that their properties can be explained by postulating a CP- ν P structure lacking the TP-layer with ellipsis of the ν P-part. On the other hand, it is shown that the properties of the cases with a pseudosluicing structure can be captured by assuming that a bare ν P lacking the CP- and TP-layers can serve as the subject of a cleft sentence, followed by AE applied to the ν P. It is also shown that the idea that the CP-layer is absent in the latter case naturally accommodate the puzzle hidden under the pseudosluicing analysis, namely the categorial mismatch between the CP-ellipsis and the TP-antecedent, together with the otherwise quite mysterious behavior of the copula in the pair of the infinitival antecedent and the past tense finite sluice. The next section argues that this kind of “smaller” clausal AE, namely the one targeting a ν P behaving as a clausal argument instead of a full-fledged CP, is independently required in order to explain the observations concerning modal mismatches.

4. Modal Mismatches under Clausal Argument Ellipsis²⁵

In this section, it is first observed that AE targeting a clausal complement (clausal argument ellipsis, CAE) allows mismatches with respect to the particles denoting certain kind of modality. That is, CAE allows modal mismatches (see also Landau 2023 for Hebrew). Then, I argue that the observation can be straightforwardly captured by assuming that CAE can target a ν P instead of a CP complement,

²⁵ The examples and the discussion to be presented in this section have been further elaborated in Takita (2024), which deals with two more modal particles in addition to the ones employed in this section. Furthermore, it is also discussed that clausal AE is insensitive to CP-level mismatches (such as declarative-interrogative mismatches), based on what Takita (2018) calls antecedent-contained clausal argument ellipsis (see also Takahashi 2024).

which is consistent with the general idea of this paper and in fact provides independent support for the analysis concerning the pseudosluicing approach proposed in the previous section.

CAE is one of the crucial components of Saito's (2004) pseudosluicing analysis. The initial evidence for CAE originates from the observation that missing clausal arguments pattern with NP and PP arguments in that they can assume sloppy reading under appropriate contexts. (see also Tanaka 2008, Takita 2010, among others). For instance, taking (31a) as the antecedent, the missing CP in (31b) can be ambiguous between the strict reading and the sloppy reading (adapted from Takita 2018:3).

(31) a. Mary-wa₁ [_{CP} zibun-no₁ iken-ga tadasii to] omotteiru

M.-Top self-Gen idea-Nom correct C think

‘(lit.) Mary₁ thinks [that self₁'s idea is correct].’

b. John-wa₂ Δ omotteinai

J.-Top not.think

‘(lit.) John₂ does not think Δ.’ (Δ = [that self_{1/2}'s idea is correct])

A stronger piece of evidence for the elided structure in missing CP arguments, hence for the idea of CAE, comes from extraction, first pointed out by Takahashi (2012) and further elaborated by Sakamoto (2017, 2020) and Takahashi (2020). Let us consider the examples in (32), which are adapted from Takita (2018:6). In both (32a-b), the sentences have the form of a cleft sentence, whose CP-subject has embedded another CP. In (32a), the operator coindexed with the focused PP *kono biru-kara* ‘from this building’ has extracted from the deeply embedded CP within the presuppositional CP-subject. Taking (32a) as the antecedent, (32b) is grammatical, where the operator corresponding to the focused PP *ano biru-kara* ‘that building’ should have extracted out of the missing CP.

(32) a. [_{CP} Op₁ Hanako-ga [_{CP} Taroo-ga t₁ detekita to] syoogensita no]-wa

H.-Nom T.-Nom came.out C testified C-Top

kono biru-kara₁ desu

this building-from Cop

‘It was from this building₁ [that Hanako testified [that Taroo came out t_1]].’

- b. [_{CP} Op₂ Yumi-ga [_{CP} ~~Taroo-ga t_2 detekita to~~] syoogensita no]-wa

Y.-Nom T.-Nom came.out C testified C-Top

ano biru-kara₂ desu

that building-from Cop

‘It was from that building₂ [that Yumi testified [that Taroo came out t_2]].’

If there were no elided structure in (32b), the operator should have no place from which it is extracted. Hence, the grammaticality of (32b) is taken as a strong indication of the presence of the hidden structure subject to CAE. In what follows, it is observed that CAE is possible even when there seems to be a mismatch between the ellipsis site and its antecedent, employing the structure of (32) as the template for the examples to be provided.

The mismatch in question has to do with certain kinds of particles denoting modality. To be more specific, I employ two modal particles out of the four particles discussed in Matsuda (2021). The first one is the particle *-tai*, which is called the optative particle (Opt). This morpheme usually serves as an adjective meaning *want* in English. As shown in (33a), it exhibits a regular tense inflection if used in a matrix clause or in an embedded clause of verbs like *omotteiru* ‘think’. In its optative usage, however, it fails to assume the past tense form. This is shown by the example in (33b), where the particle appears in a clause embedded under verbs like *nozonda* ‘hoped’ and *negatta* ‘wished’, which express a certain kind of the subject’s desire. In addition to this, these verbs do not take embedded clauses with a verb in simple present or past tense forms, as shown in (33c).

- (33) a. Miku-wa hakubutukan-e iki-tai/takatta (to omotteiru)

M.-Top museum-to go-want/wanted C think

‘Miku (thinks that she) wants/wanted to go to the museum.’

- b. Miku-wa [PRO hakubutukan-e iki-tai/*takatta to] nozonda/negatta

M.-Top museum-to go-Opt/wanted C hoped/wished

‘Miku hoped/wished [to go to the museum].’

c. *Miku-wa [PRO hakubutukan-e ik-u/ta to] nozonda/negatta

M.-Top museum-to go-Pres/Past C hoped/wished

‘Miku hoped/wished [to go to the museum].’

This property, taken together with the lack of tense-alternation exhibited by (33b), indicates that the verbs in question require obligatory control complements.

The other particle to be compared with the optative particle is what Matsuda (2021) calls the intensive particle (Int) *-(y)oo*. In fact, this is one of the several usages of the infinitive marker *-(y)oo* contained in the examples discussed in the previous sections, so that the embedded clause containing this particle is also an obligatory control complement. What is crucial for the present purpose is that the context where the intensive particle can appear blocks the optative particle as shown in (34a), and the context where the optative particle can appear blocks the intensive particle as shown in (34b).²⁶

(34) a. Miku-wa [PRO hakubutukan-e ik-oo/*iki-tai to] kimeta

M.-Top museum-to go-Int/go-Opt C decided

‘Miku decided [to go to the museum].’

b. Miku-wa [PRO hakubutukan-e *ik-oo/iki-tai to] nozonda

M.-Top museum-to go-Int/go-Opt C hoped

‘Miku hoped [to go to the museum].’

That is, the verb form *ik-oo* ‘go-Int’ but not the one *iki-tai* ‘go-Opt’ is possible in a clause embedded under verbs like *kimeta* ‘decided’, which express the speaker’s intention, while the opposite pattern is observed for clauses embedded under verbs like *nozonda* ‘hope’, which is only compatible with the optative particle.

²⁶ Here I intentionally avoid using the term *selection* to refer to the relation between the particle and the embedding verb. See footnote 29 for a related discussion.

Having laid out the necessary components of the crucial data of this section, let us start with the examples in (35), which instantiate the intensive-antecedent/optative-CAE case. (35a) is the antecedent where the intensive particle appears in the embedded clause and the embedding verb is *kimeta* ‘decided’. As is expected, when the embedding verb of the subsequent clause is *nozonda* ‘hoped’, only the optative particle is allowed, and more crucially the intensive particle is blocked, as shown in (35b). Hence, there are two obligatory control complements inside the CP-subjects in (35a-b), but they minimally differ in terms of the modal particles. Then, applying CAE to the control complements yields (35c).

(35) a. [CP Op₁ Miku-ga [CP PRO *t*₁ ik-oo to] kimeta no]-wa hakubutukan-e₁ da-ga, ...

M.-Nom go-Int C decide C-Top museum-to Cop-but

‘It is to the museum₁ [that Miku decided [to go *t*₁]], but ...’

b. [CP Op₂ Tenma-ga [CP PRO *t*₂ iki-tai/*ik-oo to] nozonda no]-wa doobutuen-e₂ da

T.-Nom go-Opt/go-Int C hoped C-Top zoo-to Cop

‘It is to the zoo₂ [that Tenma hoped [to go *t*₂]].’

c. [Tenma-ga Δ nozonda no]-wa doobutuen-e da

T.-Nom hoped C-Top zoo-to Cop

‘It is to the zoo [that Tenma hoped Δ].’

The grammaticality of (35c) indicates that CAE is legitimate even with the modal mismatches.

The same result is observed for the optative-antecedent/intensive-CAE paradigm given in (36). The antecedent (36a) contains the optative particle and the corresponding matrix verb *nozonda* ‘hoped’. The baseline example (36b) without ellipsis blocks the optative particle due to the presence of the embedding verb *kimeta* ‘decided’, leaving the intensive particle as the only possibility. Hence, the grammaticality of (36c) obtained by CAE again shows that the mismatch in question can be ignored.

(36) a. [Op₁ Miku-ga [CP PRO *t*₁ iki-tai to] nozonda no]-wa hakubutukan-e₁ da-ga, ...

- M.-Nom go-Opt C hoped C-Top museum-to Cop-but
‘It is to the museum₁ [that Miku hoped [to go t_1]], but ...’
- b. [Op₂ Tenma-ga [CP PRO t_2 ik-oo/*iki-tai to] kimeta no]-wa doobutuen-e₂ da
T.-Nom go-Int/go-Opt C decided C-Top zoo-to Cop
‘It is to the zoo₂ [that Tenma decided [to go t_2]].’
- c. [Tenma-ga Δ kimeta no]-wa doobutuen-e da
T.-Nom decided C-Top zoo-to Cop
‘It is to the zoo [that Tenma decided Δ].’

Since all the examples in (35) and (36) have the structure of cleft sentences involving ellipsis, it is ensured that the missing part has a certain syntactic structure that can accommodate extraction. Hence, it follows that the paradigms in (35) and (36) suggest that CAE allows the modal mismatches.

Let us now turn to how the observations can be captured. Note that the “mismatch” found in (35) and (36) becomes inevitable if we assume that their structure is something like (37a), where the CP argument containing the modal particle is subject to CAE.²⁷ According to the general proposal of this paper, however, the pattern in question can be analyzed in the way depicted in (37b). In this structure, a ν P directly serving as a clausal argument of the verbs in question is elided via CAE under identity with the ν P in the antecedent.²⁸

²⁷ The assumption that the modal particles occupy the position notated as T_{Mod(ality)} is just a reflection of the observation that they form a control complement, appear in between a verb stem and a complementizer, and are in complimentary distribution with the regular present and past tense morphemes. Their exact categorial status does not affect the discussion in the main text.

²⁸ Another possibility that is consistent with the general idea of this paper is illustrated in (i), where the ellipsis site contains a CP-subject which directly takes a ν P complement, lacking only the T_{ModP}. Although ellipsis may circumvent the problems arising from this structure in the same way as the one argued in Section 3, there is no way to apply CAE to obtain the intended surface string.

(i) Antecedent: ... [CP [T_{ModP} PRO₁ [ν P t_1 ... V- ν] [T_{Mod} Opt/Int]] C] V_{hope/decide} ...

- (37) a. Antecedent: ... $\boxed{[CP [T_{ModP} PRO_1 [vP t_1 \dots V-v] [T_{Mod} Opt/Int]] C]} V_{hope/decide} \dots$
 Ellipsis: ... $[CP [T_{ModP} \cancel{PRO_1} [vP \cancel{t_1} \dots V-v] [T_{Mod} \cancel{Opt/Int]] C]} V_{hope/decide} \dots$
- b. Antecedent: ... $[CP [T_{ModP} PRO_1 \boxed{[vP t_1 \dots V-v]} [T_{Mod} Opt/Int] C] V_{hope/decide} \dots$
 Ellipsis: ... $[vP \cancel{PRO_1} \dots V-v] V_{hope/decide} \dots$

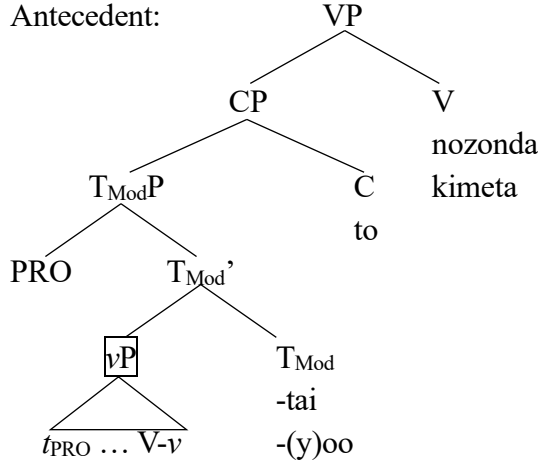
Applied to the examples at hand, (38a) and (38b) correspond to (35) and (36), respectively, and (38c) illustrates their partial structures in terms of tree diagram.

- (38) a. $[CP Op_1 \text{ Miku-ga } [CP [T_{ModP} PRO \boxed{[vP t_{PRO} t_1 \text{ ik}]] [T_{Mod} -oo]] \text{ to}] \text{ kimeta no}]-\text{wa}$
 M.-Nom go -Int C decided C-Top
 hakubutukan-e₁ da-ga, $[CP Op_2 \text{ Tenma-ga } [vP \cancel{PRO} \cancel{t_2} \cancel{\text{ik}}] \text{ nozondano}]-\text{wa}$ doobutuen-e₂ da
 museum-to Cop-but T.-Nom go hoped C-Top zoo-to Cop
 ‘(lit.) It is to the museum₁ [that Miku decided [PRO to $\boxed{[vP t_{PRO} go t_1]}]$], but it is to the zoo₂ [that Tenma hoped $[vP \cancel{PRO} \cancel{go} \cancel{t_2}]]$.’
- b. $[CP Op_1 \text{ Miku-ga } [CP [T_{ModP} PRO \boxed{[vP t_{PRO} t_1 \text{ ik}(i)]] [T_{Mod} -tai]] \text{ to}] \text{ nozonda no}]-\text{wa}$
 M.-Nom go -Int C hoped C-Top
 hakubutukan-e₁ da-ga, $[CP Op_2 \text{ Tenma-ga } [vP \cancel{PRO} \cancel{t_2} \cancel{\text{ik}}] \text{ kimeta no}]-\text{wa}$ doobutuen-e₂ da
 museum-to Cop-but T.-Nom go decided C-Top zoo-to Cop
 ‘(lit.) It is to the museum₁ [that Miku hope [PRO to $\boxed{[vP t_{PRO} go t_1]}]$], but it is to the zoo₂ [that Tenma decided $[vP \cancel{PRO} \cancel{go} \cancel{t_2}]]$.’

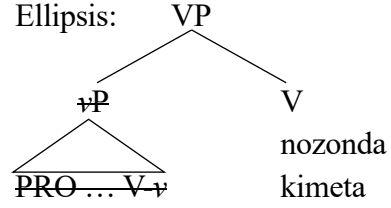
Ellipsis: ... $[CP [vP PRO_1 \dots V-v] C] V_{hope/decide} \dots$

First, applying CAE to the CP with the direct vP complement is not possible because the CP in the antecedent has a CP-T_{Mod}P-vP structure, which is not identical to the one in question. Second, eliding the vP in the ellipsis clause is not possible either. This is because it does not count as an argument of the matrix verb (as being intervened by the CP-layer) and it leaves the complementizer *to* overt, which is not the case in the relevant examples.

c. Antecedent:



Ellipsis:



In this way, the “mismatch” problem with respect to the modal particles disappears; there is no mismatch to begin with for the constituent elided by AE and its antecedent, since both of them are v Ps. This analysis also provides an independent argument for the idea developed at the end of Section 3 that a v P instead of a full-fledged CP can serve the subject of a cleft construction and applying CAE to the v P yields a pseudosluicing structure.²⁹

To summarize, this section has offered a set of novel data indicating that CAE can ignore the mismatches between the ellipsis site and its antecedent with respect to the modal particles. It is then

²⁹ I assume that the modal interpretations that seem to be available for the missing embedded clauses in (35c) and (36c) are in fact the ones inferred from the embedding verbs, because what is elided in this case is a bare v P, which is underspecified in terms of modality. Although I refer the readers to Takita (2024) for a detailed discussion, I provide one example that clearly indicates this point in (i).

- (i) Ichi-ga [_{CP} PRO soko-e iku koto]-o nozonda/kimeta
 I.-Nom there-to go fact-Acc hoped/decided
 ‘Ichi hoped/decided [to go to there].’

In (i), the clause embedded under the verbs *nozonda* ‘wished’ and *kimeta* ‘decided’ contains neither the optative nor the intensive particle, but the sentence is grammatical. This indicates that the relation between the embedding verbs and the modal particle is not the one of strict one-to-one selectional relations. In other words, the ungrammaticality of examples like (34) is due to the semantic incompatibility of the embedding verb and the embedded clause containing the modal particles.

proposed that the general idea of this paper can capture the observation rather naturally by assuming that verbs can take a ν P instead of a full-fledged CP as its complement and CAE can target such a ν P. It is also discussed that this analysis constitutes independent support for the solution to the puzzle concerning the pseudosluicing analysis, which should otherwise admit that a TP can serve as an antecedent for ellipsis of a CP.

4. Conclusion

This paper has first provided a novel set of data indicating that sluicing in Japanese allows the mismatches concerning finiteness. It is then proposed that the observation can be straightforwardly captured by assuming that sluicing can target a ν P serving as a direct complement of C instead of a TP, and that the problems arising from such an unusual complementation are essentially PF-related so that ellipsis can repair them. It is also discussed that the approach explored in this paper offers a solution to the puzzle hidden under the pseudosluicing approach to sluicing where a CP-subject is assumed to be elidable with a TP antecedent. This is because the proposed analysis makes it possible for ellipsis to target a ν P serving as the subject of a cleft sentence, taking an identical ν P as its antecedent. Finally, it is shown that the idea receives independent support from the analysis of the other set of novel data concerning mismatches with respect to the modal particles in a clause embedded under a certain kind of verbs.

There are several remaining issues, both empirically and theoretically. As mentioned at the outset of this paper, recent studies on sluicing have shown that it is insensitive to various kinds of mismatches around and above the TP-layer such as finiteness, tense, modality and polarity. Although this paper has shown that sluicing and clausal argument ellipsis in Japanese are possible even with the mismatches regarding finiteness and the ones with a limited kind of modal elements, respectively, it is clear that much more work is necessary to complete the whole picture. In fact, given the proposed analysis where seemingly CP/TP-ellipsis can be analyzed as ν P-ellipsis, it is predicted that any kind

of mismatches at that level are allowed. As for the theoretical point, one issue arising from the proposed analysis is why the TP (and above) can be skipped but not below. That is, if it were possible for a VP lacking ν P (or any functional projection responsible for voice) to serve as a direct complement of C and to be elided by sluicing, it would be incorrectly predicted that sluicing can ignore voice-mismatches. Although at this point I have to leave these important issues left unexplored, I believe that the contribution of this paper is significant enough because it provides a step toward tackling the question of what kind of mismatches are allowed in ellipsis and why.

Acknowledgments: I am grateful to the participants of the workshop “Current Issues in Comparative Syntax 2: Boundaries of Ellipsis Mismatch” (Tsuda University, Tokyo/Hybrid, September 2023) and especially to Shun Ihara, Taichi Nakamura, Yuta Sakamoto, and Yosuke Sato for their helpful comments and suggestions. Special thanks go to two anonymous reviewers and the editors of *JEAL* for their valuable feedback. Some portions of this paper were also presented at The 17th Annual Meeting of The Kansai Branch of The English Literary Society of Japan (Konan University, Kobe, December 2022), 10th Okayama Linguistics Forum (Norte Dame Seishin University, Okayama, February 2023), and Comparative Syntax, Semantics and Language Acquisition #2 (Nanzan University, Nagoya, January 2024). I would like thank the audiences at these events for their valuable feedback, including Yoshihito Dobashi, Tetsuro Hiratsuka, Yuto Hirayama, Katsumasa Ito, Hideki Kishimoto, Masako Maeda, Kenta Mizutani, Nozomi Moritake, Jun Onume, Mamoru Saito, Mizuki Sakamoto, Koji Sugisaki, Hidekazu Tanaka, and Masashi Yamaguchi. Part of this work is supported by JSPS KAKENHI Grant Number 23K00581 (PI: Kensuke Takita) and 21H00532 (PI: Hiroki Narita). All errors are of course mine.

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